

Team Worksheet — Lab 9

Data Types, Operational Definitions and the Data Collection Plan

Team members: _____

Date: _____

1. Our understanding of the problem

2. Our working

Use the data in `lab-09-data-collection-raw-extract.xlsx`.

3. What we produced

Deliverable: A data collection plan with operational definitions for every metric in your capstone.

4. What the evidence does NOT tell us

5. Our conclusion

6. Step notes

Step 1. Recap the data hierarchy in full: nominal, ordinal, interval and ratio — and why the type determines which statistical test is legal.

Step 2. Recap continuous versus discrete (attribute) data, and why continuous data needs far smaller samples to reach the same confidence.

Step 3. List every metric your capstone requires: the project Y, each sub-Y from your flowdown, and each candidate X you intend to test.

Step 4. Classify each metric by data type. Flag any metric currently collected as attribute that could be captured as continuous — convert it if you can.

Step 5. Write a complete operational definition for each metric: what is measured, the instrument, the unit, the sampling point, the collector and the defect criterion.

Step 6. Test each operational definition by having two people apply it independently to the same item. If they disagree, the definition is not yet operational.

Step 7. Identify your stratification factors — plant, line, shift, operator, product family, supplier. You must capture these at collection time or you can never stratify later.

Step 8. Build the data collection plan table: metric, definition, type, source, sample size, frequency, collector, stratification fields and start date.

Step 9. Design the check sheet or data form. Make the easy path the correct path, or collectors will improvise.

Step 10. Dry-run the plan on a small batch and fix whatever breaks before committing to full collection.

7. Self-check

Every capstone metric has a written operational definition that two people apply identically, stratification fields are captured at source, and the plan has survived a dry run.

Complete? ■ Yes ■ Not yet

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